

DollarPunk

*ETL-NLP for Sentiment Monitoring and Decision Support in Behavioral
Finance*

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1 Motivation

During my professional career, I worked for three years in IT consulting in the field of data engineering, focusing on organizing and managing data flows through ETL processes for some of the largest Italian companies. I had the opportunity to see firsthand which data is used and often had to migrate code handling such data from on-premises platforms to cloud platforms.

This experience allowed me to closely observe how complex these systems can become. Starting from these experiences, I began asking myself a few questions:

In which cases can technology truly be pushed to its limits?

How can unstructured data, such as media news or social media posts, be used to generate value?

This research proposal aims to develop an application, named *DollarPunk*, that assists users in monitoring the digital sentiment related to a list of financial securities.

2 State of the Art

2.1 Pioneering studies

One of the pioneering studies in *sentiment analysis* is that of Tetlock (2007).

In particular, Tetlock quantified the “pessimism” of articles in the Wall Street Journal and found that high levels of negative tone in the press anticipate downward pressure on stock prices, followed by a return to market fundamentals.

This early result provided evidence that sentiment extracted from texts—such as the pessimistic or optimistic tone of news—can influence market behavior.

2.2 Social Media and Investor Sentiment

With the rise of social media and online platforms in the 2010s, the sources of sentiment data have expanded. Investors—especially retail investors—began expressing opinions on financial forums, Twitter, and blogs, providing a stream of text rich with real-time information about market sentiment.

Literature from this period has examined the extent to which sentiment influences stock returns, often in contrast with traditional financial theory, which excludes emotional factors. Zachary McGurk and Hall (2020) clearly show this paradigm shift: by analyzing social media posts, the authors construct a quantitative measure of investor sentiment and find that this measure has a positive and significant effect on abnormal stock returns.

An emblematic case that catalyzed interest in sentiment analysis was the COVID-19 pandemic. Yuejiao Duan (2021) *et al.*, for example, developed COVID-19-related sentiment indices using millions of texts from both official media and social media in China.

2.3 The Rise of Transformer Models and FinBERT

In the late 2010s, the introduction of BERT (Devlin et al. (2018)) revolutionized NLP, including in the financial domain, thanks to high-quality contextual language representations. However, adapting to the technical language of finance required specialized models. FinBERT, proposed by Araci (2019), was the first model pre-trained on financial texts and then fine-tuned for sentiment analysis, outperforming previous methods by about 15 percentage points on datasets such as the *Financial PhraseBank*.

Later versions, such as the one by Huang (2023), trained on large financial corpora, achieved up to 89.7% accuracy in detecting negative sentences, improving the understanding of linguistic nuances compared to generic models.

FinBERT is now a standard for financial NLP tasks: tone analysis in reports, news classification, and information extraction from financial statements. Gu et al. (2024) show that a FinBERT+LSTM model improves stock trend forecasting compared to traditional LSTMs, highlighting the value of domain specialization.

Finally, the emergence of specific benchmarks (e.g., FiQA, Financial PhraseBank) has supported comparative evaluations, while recent extensions focus on entity-level analysis Yixuan Tang et al. (2023).

2.4 General vs. Domain-Specific Language Models in Finance

In 2022–2023, attention shifted to general-purpose LLMs such as ChatGPT (GPT-3.5/4), which can interpret financial texts despite lacking domain-specific training. Lopez-Lira and Yuehua Tang (2024) show that ChatGPT can predict stock price direction based on news headlines. Even in zero-shot settings, models like GPT-3.5/4 can perform sentiment analysis on financial documents, though with limitations regarding specialized contexts or recent events (Shobayo et al. (2024)).

These results have driven interest in specialized LLMs. BloombergGPT Wu et al. (2023), with 50 billion parameters and a 363-billion-token financial corpus, is the first large-scale LLM dedicated to finance, offering improved accuracy on industry-specific prompts thanks to its deep domain knowledge.

In parallel, the open-source FinGPT project Yang, Liu, and C. D. Wang (2023) aims to make financial LLMs accessible by fine-tuning open-source models on public financial data. With a data-centric approach, FinGPT enables sentiment analysis, automatic report generation, and low-code development, while operating with fewer resources than proprietary models.

2.5 Recent Developments and Trends from 2024 Onward

Despite progress, challenges remain in NLP for financial sentiment. Shobayo et al. (2024) show that a simple logistic regression model outperformed both FinBERT and GPT-4 in accuracy, indicating that greater complexity does not always lead to better performance.

New metrics are expanding the scope: Cao, Wunkaew, and Geman (2025) propose the Hype Index, which measures media attention rather than tone. By analyzing the S&P 100, they demonstrate that spikes in coverage can anticipate volatility and market movements, highlighting the predictive value of “information intensity.”

In summary, from 2000 to 2025, NLP models have evolved from simple analyses to advanced LLMs. Current trends focus on explainable AI, models integrated into real-time

pipelines, and analyses that consider not only sentiment but also attention, emotions, and complex textual dynamics—making the field increasingly rich and mature.

3 Project Description

3.1 Development Tools, Versioning, and Hosting

The project adopts a fast and experimental approach, supported by AI tools for refactoring, debugging, and code generation, under human supervision. Code will be managed via GitHub with strategic checkpoints.

Python, Rust, or JavaScript will be used depending on the module, with a containerized architecture (e.g., Docker) to ensure portability. Server/API hosting will be evaluated between PaaS and VPS solutions, balancing cost and control. For compute-intensive tasks or training, cloud services such as Databricks or RunPod will be used.

The setup remains flexible and open to changes during development. A demo is available at: ***Financial News Collection Demo***

3.2 Data Collection

Forecasting market movements requires acquiring large volumes of textual data from heterogeneous sources:

- **Financial sources:** Bloomberg, Reuters, WSJ, Yahoo Finance, ANSA, Il Sole 24 Ore, Investing.com, etc.
- **Social media and communities:** Twitter, Reddit (*r/investing*, *r/wallstreetbets*), LinkedIn, Telegram, YouTube.
- **Search engines:** SerpApi, duckduckgo-search, ddgs.
- **Topics of interest:** financial sentiment, macroeconomic trends, geopolitical events, earning reports, ESG indicators, cryptocurrencies.

Data acquisition will occur via scraping (e.g., the **newspaper** library) or via APIs when available. On Reddit, collecting comments, upvotes, and social metrics will also be relevant. For all channels, a strategy will be defined to structure and contextualize the data consistently.

3.3 Text Data Analysis

Once a sufficient amount of textual data has been collected, it will be useful to develop a function that processes the text (whether a news article or a post) and returns a JSON array with key information: mentioned tickers, *relevance score*, and *sentiment score*.

Example of Expected Output:

```
1 [
2 {
3 "title": "If You Invested $1,000 In Lululemon Stock In 2013, Here's
4 How Much You'd Have Now",
5 "url": "https://www.fool.com/investing/2023/04/30/if-you-invested-1
6 000-in-lululemon-stock-in-2013-he/",
7 "source": "Motley Fool",
8 "time_published": "2023-04-30T13:34:00",
9 "stocks_mentioned": ["LULU", "NKE"],
10 "ticker_sentiments": [
11 {
12 "ticker": "LULU",
13 "relevance_score": 0.11,
14 "sentiment_score": 0.12,
15 "sentiment": "Neutral"
16 },
17 {
18 "ticker": "NKE",
19 "relevance_score": 0.27,
20 "sentiment_score": 0.22,
21 "sentiment": "Somewhat-Bullish"
22 }
23 ]
24 ]
```

Various solutions will be proposed and evaluated to reliably extract and compute the three fields (*tickers*, *relevance score*, *sentiment score*). Given the highly dynamic and evolving nature of NLP applied to finance, more effective approaches or tools may emerge even in the short term, and will be adopted if deemed advantageous.

3.3.1 Tickers

To associate a text with the correct ticker, multiple signals should be combined: context of the query (e.g., targeted queries like "Apple stock"), Named Entity Recognition (e.g., "Apple Inc."), pattern matching (e.g., \$AAPL, (NASDAQ:AAPL)), and dictionaries mapping company names to tickers.

A robust system integrates these sources and assigns a *confidence score*: high (0.9–1.0) for direct citations or explicit queries, medium (0.6–0.8) for company names, lower for indirect mentions. This reduces ambiguity and improves accuracy in linking text to stock symbols.

3.3.2 Relevance Score

The *relevance score* measures how central a ticker is to a given text. A simple method is to count occurrences and normalize them by length or number of total tickers.

More advanced approaches include using attention weights in Transformer models, reinterpreted as relevance indicators Corro and Hoffart (2021), and commercial systems

like RavenPack, which evaluate frequency, position, and semantic role. The News-to-Numbers framework (Stanford NLP, 2023) assigns scores from 0–100 to ticker–headline pairs using LLMs (Group 2023).

In short, basic counting is useful for prototyping, but professional applications require more contextual and automated methods.

3.3.3 Sentiment Score

The *sentiment score* quantifies the tone associated with a ticker in the text, mapping polarity (positive, negative, neutral) onto continuous scales (e.g., -1 to $+1$), using models like FinBERT or specialized LLMs.

For greater precision, entity-level sentiment (ticker–sentence) is used (Yixuan Tang et al. 2023), as done by commercial systems (Refinitiv, Bloomberg) and open-source ones (FinGPT (Yang, Liu, and C. D. Wang 2023)).

Alternatively, zero-shot models like GPT-4 can generate continuous scores (e.g., 0–100) via prompts, without specific training (Shobayo et al. 2024), making them useful for automated, flexible pipelines.

3.3.4 Fine-Tuning

Fine-tuning an open-source LLM allows building a model that, given a text, directly returns a structured object with *ticker*, *relevance score*, and *sentiment score*. This approach unifies the pipeline and avoids separate modules, leveraging the generalization of the pre-trained model.

Models like *FinBERT* (Araci 2019) or *Mixtral*, as well as adaptable open-source LLMs (e.g., *FinGPT* (Yang, Liu, and C. D. Wang 2023)), can be fine-tuned using annotated financial data. For this, a large dataset with entity-level labels (ticker-level sentiment) is essential, such as those from public challenges (Maia, Almeida, and Gonçalves 2018) or aggregated via news APIs like Alpha Vantage.

Training can be done on GPU-enabled cloud infrastructure, offered by services like RunPod, Lambda Labs, or Google Cloud, enabling flexible cost-performance trade-offs (Li, Zhao, and X. Wang 2020).

3.3.5 Retrieval-Augmented Generation (RAG)

RAG combines LLMs (e.g., GPT-4, Mixtral) with a document retrieval module: the model generates responses based on real-time extracted context from a vector index (e.g., financial news or headlines).

The input can be a standardized prompt (“Extract ticker, sentiment score, and relevance score”), and the model directly outputs structured JSON without fine-tuning. The system remains updatable by modifying the indexed documents (Lewis et al. 2020).

Tools like FAISS and LangChain enable a scalable, flexible pipeline, as shown in projects like FinGPT-RAG (Gudibande et al. 2023; Yang, Liu, and C. D. Wang 2023).

3.3.6 Summary

Given a labeled dataset (ticker-level sentiment), several strategies are possible:

- **FinBERT fine-tuning:** using GPUs (e.g., RunPod) to refine FinBERT to directly generate JSON with sentiment scores per ticker via supervised learning.

- **Fine-tuning of generic open-source LLMs:** models like Mixtral or FinGPT can be adapted via labeled datasets, also requiring powerful GPUs.
- **RAG approach:** if GPU resources are limited, index the dataset and use a retrieval-based LLM (e.g., GPT-4, Mixtral) to generate JSON from retrieved context, avoiding fine-tuning (see Lewis et al. 2020).

If no labeled dataset is available:

- **Baseline model with FinBERT:** apply zero-shot sentiment classification on sentences to generate initial scores.
- **GPT-4 via API:** can generate structured JSON, but may be costly at scale.
- **Manual dataset construction:** manually annotate examples, useful for subsequent fine-tuning even with limited resources.

3.4 Temporal Decay (on Relevance Only)

Once *tickers*, *sentiment*, and *relevance* have been calculated, it is useful to distribute them over time to obtain an aggregated daily view. To do so, we propose applying temporal decay to the *relevance score* only: sentiment remains unchanged but is weighted based on the residual relevance of the news item in the days following its publication.

The goal is to construct, for each day, a synthetic measure of active market sentiment, accounting for the fact that the informational impact of news tends to decrease over time.

Example

In the table below, the relevance score decays each day, while the associated sentiment remains fixed:

News #	Day 0	Day 1	Day 2
1	S=1.000, R=0.800	S=1.000, R=0.593	S=1.000, R=0.442
2	—	S=0.900, R=0.700	S=0.900, R=0.518
3	S=0.750, R=0.600	S=0.750, R=0.444	S=0.750, R=0.329
Weighted Avg.	≈ 0.892	≈ 0.716	≈ 0.697

Relevance is decayed using an exponential function:

$$R_{i,d} = R_{i,0} \cdot e^{-\lambda \Delta t}, \quad S_{i,d} = S_{i,0}$$

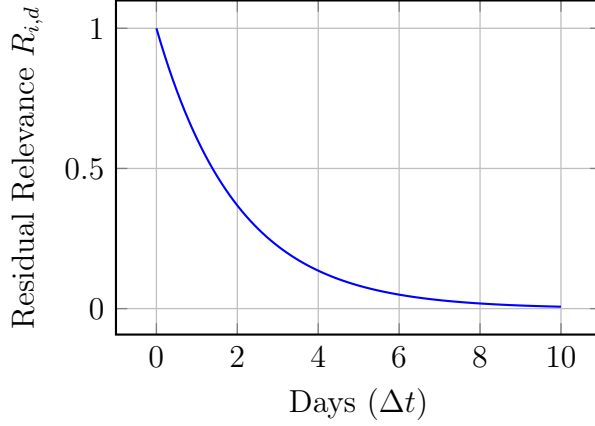


Figure 1: Exponential decay of the relevance score with $\lambda = 0.5$

Where:

- $R_{i,0}$ and $S_{i,0}$ are the initial relevance and sentiment values of news item i
- Δt is the number of days since publication
- λ is a decay parameter to be estimated

For each day d , the sentiment is averaged and weighted by relevance:

$$\text{AvgW}_d = \frac{\sum_i (S_{i,0} \cdot R_{i,d})}{\sum_i R_{i,d}}$$

This approach aligns with the logic adopted by providers like RavenPack, which apply exponential decay to the relevance score to model informational relevance over time (RavenPack 2025c; RavenPack 2025b; RavenPack 2025a).

3.4.1 Future Developments

In the previous model, each news item’s relevance score is decayed over time according to the exponential function:

$$R(t) = R_0 \cdot e^{-\lambda t}$$

This function is the solution to the ordinary differential equation:

$$\frac{dR(t)}{dt} = -\lambda R(t)$$

This equation defines a rule governing the continuous decay of relevance over time, consistent with methods used by various providers to model the fading impact of information.

However, this approach assumes that each news item acts independently. In reality, many articles cover similar topics, refer to related events, or share a common semantic context. Treating them as isolated risks ignoring broad or correlated informational effects.

A potential future development is to extend the temporal decay of the relevance score by introducing a semantic diffusion term. In other words, a news item’s relevance could not only decay over time but also propagate to semantically related content.

This leads to a generalization of the decay equation:

$$\frac{\partial R(x, t)}{\partial t} = D \nabla^2 R(x, t) - \lambda R(x, t)$$

Where:

- $R(x, t)$ is relevance as a function of time t and position x in semantic space (e.g., text embeddings),
- D is the semantic diffusion coefficient,
- λ is the temporal decay rate.

In this model, the informational relevance of one news item can spread to similar news, modeling a more realistic dynamic of the information flow. Sentiment remains unchanged but is weighted by $R(x, t)$, which evolves in both time and semantic space.

The main challenge is defining a robust semantic distance between texts, using embeddings, similarity networks, or clustering. In a coherent semantic space, the relevance score can propagate via numerical methods or graphs, enabling applications in risk monitoring, portfolio signals, and event tracking.

This allows us to conceptualize relevance not as a static score, but as a dynamic field in continuous evolution.

3.4.2 Innovation and Related Work

Modeling sentiment as a dynamic field that diffuses across related news is still underexplored. Most studies treat news as independent units or aggregate sentiment over time.

Some recent works approach this idea: Wan et al. (2021) show sentiment diffusion in information networks, while M. Wang, Cohen, and Ma (2024) introduce FININ, which considers semantic interactions among news items to predict markets.

However, the explicit use of a partial differential equation model—such as a diffusion equation with decay—is still novel in the NLP-finance domain. This continuous and interpretable formulation goes beyond discrete networks and weighted averages, enabling new applications in forecasting, portfolio strategies, and interpreting information flows.

3.5 Exploratory Analysis: Sentiment and Price

After the data collection and scoring phase, it is essential to verify whether the extracted sentiment is actually linked to price movements. Exploratory analysis provides an initial plausibility check by comparing the daily evolution of sentiment with market changes.

A simple approach involves comparative charts of sentiment and closing prices, a scatter plot with percentage price variation, and a correlation measure (Pearson or Spearman), even with time lags.

Previous studies, such as StockSnips (2016) and Cristescu et al. (2023), have found significant correlations between news sentiment and the prices of stocks like Apple, Microsoft, and Tesla.

This phase is iterative: if no signals emerge, the data or scoring method should be revisited. If a partial relationship is observed, one can move toward more sophisticated models.

3.5.1 Simple Strategy Backtesting

Backtesting involves simulating the application of a strategy on historical data to assess its effectiveness. “Back” refers to operating retrospectively: historical series of sentiment and prices are used to evaluate how a decision logic would have performed.

A simple example: enter a long position if sentiment is positive, stay out (or go short) if it’s negative or neutral. The entry occurs the day after the signal, with an exit at end of day or after n days.

Metrics to evaluate include:

- Cumulative return vs. benchmark
- Prediction accuracy
- Simplified Sharpe ratio

This is a low-complexity first operational test. If results are promising, more advanced strategies can be pursued.

3.5.2 Portfolio Simulation

To make the test more realistic, the strategy can be extended to a multi-ticker context. The simulation defines initial capital, allocation among stocks, exit rules (e.g., stop-loss), and includes trading costs. The goal is not yet optimization, but to assess whether the strategy holds under plausible conditions. If the results suggest robustness, the next phase—with predictive models like LSTM—can be approached with greater confidence.

3.5.3 Integration with LSTM Models

A possible next step is to build a predictive model that uses price data, volumes, technical indicators, and sentiment together. Several studies (StockSnips 2016; Ouf et al. 2024; Chaudhary 2025) show that combining these signals in an LSTM improves forecasting ability compared to using price data alone.

But what is an LSTM? It’s a type of neural network used for analyzing time series—data that changes over time. Here’s how it works: to make a prediction for “today,” the model looks at a “window” of data from previous days (e.g., the past 5 days). In each window, it might see the price, volume, and sentiment for each day.

In practice, the LSTM learns to recognize patterns in the past that anticipate future movements. By adding sentiment as extra input, the model can better understand market moves influenced not just by numbers but also by what “people are saying.”

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